

PROJECT AND TEAM INFORMATION

TrackMate – Smart Railway Reservation Platform

A web based platform made to simplify the process of searching and booking of railway tickets. The system will provide users a secure login facility and a simple interface for selecting their journey details. Dehradun will be set as the default location, allowing user's to select their desired local source location and destination within the available railway network. Users will be able to choose a journey date within a one month period from the current date, and view the trains available for their selected journey.

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Railway ticket reservation is an important service for students, working professionals, tourists, and other passenger who travel frequently between different locations. However, searching for suitable trains and managing reservations can sometimes be inconvenient, since user have to navigate through multiple options, enter repetitive information, or deal with additional platform and convenience charges. A simple and focused booking system can make this process more easier by bringing the essential reservation functions together in one platform. The motivation behind this project is to develop a user friendly web based Railway Reservation System that allow passengers to search available trains and manage their bookings conveniently, while keeping the interface clear and the navigation straightforward. Another important motivation is to provide a transparent booking experience without adding any platform fees to the ticket cost.

State of the Art / Current solution (1 pt)

Existing railway reservation system allow users to search trains, check schedules, view seat availability, and book or cancel tickets online, generally by entering source, destination, journey date, and passenger details. These systems provides extensive features but may be complex for users who are seeking only basic booking services. Our proposed system focus on a simpler and more user-friendly reservation process, using Dehradun as the default location and allowing train search within a one month date range, without imposing any additional platform or convenience fees.

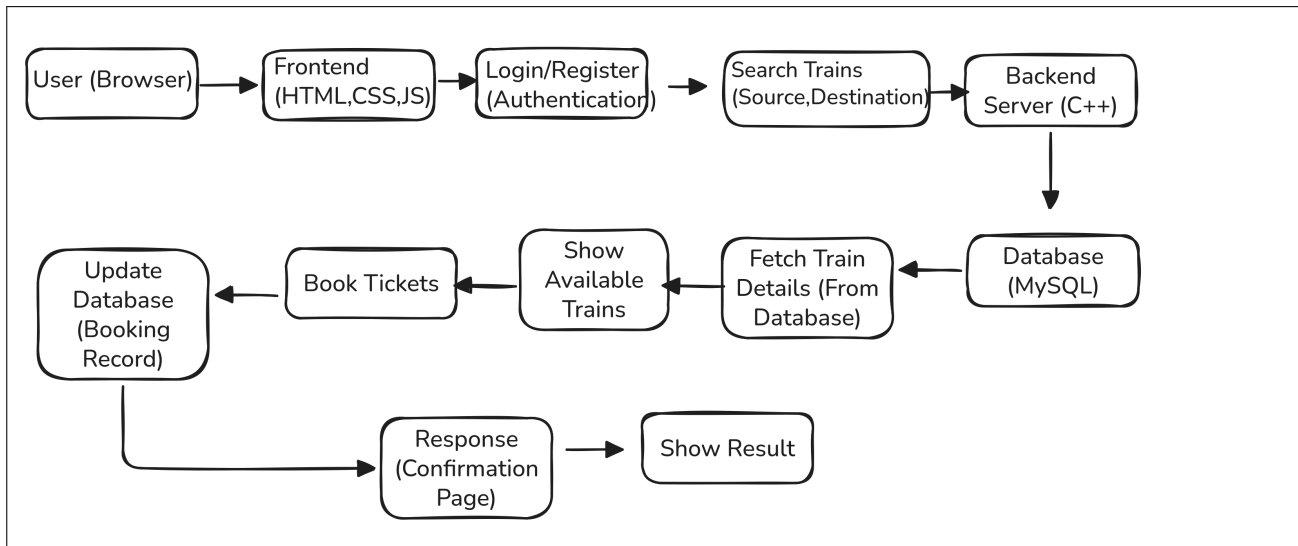
Project Goals and Milestones (2 pts)

The primary goal of this project is to develop a simple, reliable, and user-friendly web-based Railway Reservation System that make searching trains and booking tickets more convenient. The system will provide secure user login functionality and use Dehradun as the default location, allowing users to select destinations, journey dates, and preferred travel times. Based on the selected journey details, the system will retrieve and display available trains along with train number, train name, departure and arrival timings, journey duration, and seat availability. Users will also be able to enter passenger details, complete a reservation, and manage their bookings, without any platform or convenience fees. The project will begin with requirement analysis and identifying the necessary user, train, and booking information, followed by designing the interface, workflow, and database structure. The next milestone will be implementing login and authentication, followed by the train-search module based on location, date, and time. Booking and cancellation functionality will then be integrated with the database, and the complete system will undergo testing and refinement to ensure all components work together as intended.

Project Approach (3 pts)

The project will begin with requirement analysis and planning of the railway reservation workflow. Dehradun will be set as the default location, with option for selecting destinations and journey dates. The system will then be design with a simple web interface, reservation logic, and database for storing users, trains, passengers, and bookings. Finally, booking, cancellation, and booking management feature will be integrated and tested, and the system refined to ensure proper functionality, reliability, and ease of use.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The project will result in a functional web based Railway Reservation System that provide a simple and convenient platform for searching and booking train tickets, including user login and registration, train search based on source, destination and journey date, display of train timings and seat availability, and basic booking management such as viewing and cancellation of reservations. User, train, passenger, and booking information will be stored and manage through a database. The project will demonstrate the practical implementation of Data Structures, OOP with C++, DBMS, Operating Systems concepts, and web development, with the final deliverable being a working prototype with a simple interface, organized database, and complete basic reservation workflow.

Assumptions

The system is assumed to operate with Dehradun as the default location, with journey dates limited to one month from the current date. Train, user, passenger, and booking information will be maintain in the system database. The project will focus on basic railway reservation functionality and will not include real-time railway data or actual payment processing, and is intended primarily for academic and demonstration purpose.

References

- [1] Indian Railway Catering and Tourism Corporation (IRCTC), Official Railway Ticket Booking and Reservation Services, Government of India. [Online]. Available: <https://www.irctc.co.in/>. [Accessed: Sep. 9, 2026].
- [2] Oracle Corporation, MySQL 8.4 Reference Manual, MySQL Documentation. [Online]. Available: <https://dev.mysql.com/doc/refman/8.4/en/>. [Accessed: Sep. 9, 2026].
- [3] MDN Web Docs, Learn Web Development: HTML, CSS and JavaScript, Mozilla. [Online]. Available: <https://developer.mozilla.org/en-US/docs/Learn>. [Accessed: Sep. 9, 2026].
- [4] B. Stroustrup, The C++ Programming Language, 4th ed. Boston, MA, USA: Addison-Wesley Professional, 2013.